This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scripter Modules, refer to the "Guide to Using Scripter Modules" document.

Device Specifications

Device Type: Video Projector
Manufacturer: Canon
Firmware Version: N/A
Model(s): LV-HD420, LV-X420

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scripter Version
2.06.0001-b003	1.4.1

Version History

Module Version	Date	Notes
1_0_0_0	11/14/2017	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
  Example: `InterfaceName.Unidirectional = 'True'`

- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
  Example: `InterfaceName.connectionCounter = 5`

- DeviceID variable must be set accordingly. Default value is '1'. Device ID ranges from '0' - '98' and 'Broadcast'.
  Example: `InterfaceName.DeviceID = '0'`

Supported Classes and Examples

SerialClass
`InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='LV-HD420')`
SerialOverEthernetClass
`InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='LV-HD420')`
EthernetClass
`InterfaceName = ModuleName.EthernetClass('192.168.254.254', 23, Model='LV-HD420')`

Set Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command	Value	Value	Value
AspectRatio	'Fill'	'4:3'	'16:9'
	'Letter Box'	'Real'	'2.35:1'

```
# AspectRatio example
InterfaceName.Set('AspectRatio', 'Fill')
```

Command	Value
AutoImage	None

```
# AutoImage example
InterfaceName.Set('AutoImage', None)
```

Command	Value	Value
Freeze	'On'	'Off'

```
# Freeze example
InterfaceName.Set('Freeze', 'On')
```

Command	Value	Value	Value
Input	'RGB 1'	'RGB 2'	'Video'
	'S-Video'	'HDMI'	'Component'
	'HDMI 2 / MHL'		

```
# Input example
InterfaceName.Set('Input', 'RGB 1')
```

Command	Value	Value	Value
LampMode	'Normal'	'Eco'	'Smart Eco'

```
# LampMode example
InterfaceName.Set('LampMode', 'Normal')
```

Command	Value	Value	Value
Power	'On'	'Off'	'Reset'

```
# Power example
InterfaceName.Set('Power', 'On')
```

Command	Value	Value
VideoMute	'On'	'Off'

```
# VideoMute example
InterfaceName.Set('VideoMute', 'On')
```

Command	Value
Volume	0 to 10 in steps of 1

```
# Volume example
InterfaceName.Set('Volume', 10)
```

Status Available

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)
```

FeedbackHandler will be called only when the specified qualifier gets a new status.

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
```

FeedbackHandler will be called when any qualifier gets a new status.

Command	Value	Value	Value
AspectRatio	'Fill'	'4:3'	'16:9'
	'Letter Box'	'Real'	'2.35:1'

```
# AspectRatio examples
InterfaceName.Update('AspectRatio')
Value = InterfaceName.ReadStatus('AspectRatio')
InterfaceName.SubscribeStatus('AspectRatio', None, FeedbackHandler)
```

Command	Value	Value
ConnectionStatus	'Connected'	'Disconnected'

```
# ConnectionStatus examples
Value = InterfaceName.ReadStatus('ConnectionStatus')
InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)
```

Command	Value	Value
Freeze	'On'	'Off'

```
# Freeze examples
InterfaceName.Update('Freeze')
Value = InterfaceName.ReadStatus('Freeze')
InterfaceName.SubscribeStatus('Freeze', None, FeedbackHandler)
```

Command	Value	Value	Value
Input	'RGB 1'	'RGB 2'	'Video'
	'S-Video'	'HDMI'	'Component'
	'HDMI 2 / MHL'		

```
# Input examples
InterfaceName.Update('Input')
Value = InterfaceName.ReadStatus('Input')
InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)
```

Command	Value	Value	Value
LampMode	'Normal'	'Eco'	'Smart Eco'

```
    # LampMode examples
    InterfaceName.Update('LampMode')
    Value = InterfaceName.ReadStatus('LampMode')
    InterfaceName.SubscribeStatus('LampMode', None, FeedbackHandler)
```

Command	Value		
LampUsage	0 – 9999		

```
    # LampUsage examples
    InterfaceName.Update('LampUsage')
    Value = InterfaceName.ReadStatus('LampUsage')
    InterfaceName.SubscribeStatus('LampUsage', None, FeedbackHandler)
```

Command	Value	Value	Value
Power	'On'	'Off'	'Cooling'
	'Reset'		

```
    # Power examples
    InterfaceName.Update('Power')
    Value = InterfaceName.ReadStatus('Power')
    InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)
```

Command	Value	Value
VideoMute	'On'	'Off'

```
    # VideoMute examples
    InterfaceName.Update('VideoMute')
    Value = InterfaceName.ReadStatus('VideoMute')
    InterfaceName.SubscribeStatus('VideoMute', None, FeedbackHandler)
```

Command	Value
Volume	0 to 10 in steps of 1

```
    # Volume examples
    InterfaceName.Update('Volume')
    Value = InterfaceName.ReadStatus('Volume')
    InterfaceName.SubscribeStatus('Volume', None, FeedbackHandler)
```

Cable and Adapter Requirements

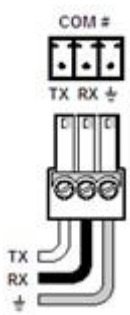
Captive Screw to Female DB9 RS-232 Serial Cable

Notes for the Device

Serial communication

Port Type:	RS-232	**Parity:**	None
Baud Rate:	9600	**Stop Bits:**	One
Data Bits:	8	**Flow Control:**	None

Pin Assignments Diagra



Please contact the manufacturer for pin diagram information.

Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scripter ethernet interface

Port Type:	Ethernet
Default Port:	23
Logon Credentials Supported:	No
Multi-Connection Capabilities:	Undetermined
Port Changeability:	Yes

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

Appendix A. Set Commands

Aspect Ratio 16:9	V00S03012\x0D
Aspect Ratio 2.35:1	V00S03015\x0D
Aspect Ratio 4:3	V00S03011\x0D
Aspect Ratio Fill	V00S03010\x0D
Aspect Ratio Letter Box	V00S03013\x0D
Aspect Ratio Real	V00S03014\x0D
Auto Image None	V00S0003\x0D
Freeze Off	V00S03040\x0D
Freeze On	V00S03041\x0D
Input Component	V00S0208\x0D
Input HDMI	V00S0206\x0D
Input HDMI 2 / MHL	V00S0209\x0D
Input RGB 1	V00S0201\x0D
Input RGB 2	V00S0202\x0D
Input S-Video	V00S0205\x0D
Input Video	V00S0204\x0D
Lamp Mode Eco	V00S03190\x0D
Lamp Mode Normal	V00S03191\x0D
Lamp Mode Smart Eco	V00S03192\x0D
Power Off	V00S0002\x0D
Power On	V00S0001\x0D
Power Reset	V00S0006\x0D
Video Mute Off	V00S03020\x0D
Video Mute On	V00S03021\x0D
Volume 0	V00S03050\x0D
Volume 10	V00S030510\x0D

Appendix B. Query Commands

Aspect Ratio	V00G0301\x0D
Freeze	V00G0304\x0D
Input	V00G0220\x0D
Lamp Mode	V00G0319\x0D
Lamp Usage	V00G0004\x0D
Power	V00G0007\x0D
Video Mute	V00G0302\x0D
Volume	V00G0305\x0D